JOSHUA HEAGLE

Senior Full-Stack Engineer

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Summary

Full-stack engineer with 5+ years designing and shipping systems in PHP/Laravel, with a consistent pattern of turning one-off solutions into shared, maintainable architecture — extracting monolithic services into composable patterns, and driving internal tooling from personal use into team-wide packages. Currently building in-house engineering capability for a company-wide supply-chain initiative, having been retained through a major business restructuring as one of a small core technical team. Maintains a substantial personal ecosystem of open-source TypeScript libraries, and has hands-on experience directing AI coding agents through complex, multi-repo engineering work.

Skills

Languages & Frameworks: PHP 8, Laravel, JavaScript, TypeScript, Vue.js, React

Architecture & Patterns: Repository Pattern, Strategy Pattern, DTOs, Event-Driven Design (Jobs/Events/Listeners/Observers), Package Design

Data & Infrastructure: MySQL, MSSQL, Docker, AWS, Elasticsearch, Algolia

Testing & Tooling: PHPUnit, Jest, Laravel Dusk, TDD, CI/CD (GitHub Actions), Custom Build Tooling

AI Tooling: Directing AI coding agents through complex, multi-repo engineering work

Experience

Full-Stack Engineer — *BIC (formerly Inkbox), Toronto, ON* April 2021 – Present

- **Selected as one of four engineers (plus a QA) retained** to build in-house engineering capability on a newly formed Global Supply Chain team, after the consumer Inkbox business closed and BIC chose to invest in an internal team rather than continue relying on third-party contractors.
- **Designed and built an Inventory & Distribution system from scratch** — data model, schema, admin tooling, and integration with the platform's DTO layer — covering a catalog of tens of thousands of SKUs; hardened it through a real production bug-fix cycle.
- **Refactored a 1,489-line monolithic sheet-generation service** into a Strategy-pattern architecture (a shared interface, six generator types, composed traits), cutting the core class to 128 lines; later extended the same pattern to support bundled products and added caching for performance.
- **Introduced the platform's Data Transfer Object pattern** and maintained it for years; drove its extraction into a standalone shared package now used across multiple codebases.
- **Built and maintained the discount system and product feed pipeline** (spanning both the core backend and the customer-facing storefront) over multiple years each.
- **Prototyped and benchmarked a sort/packing algorithm** for a roll-to-roll print-sheet generator, choosing PHP's built-in usort over a custom implementation after measuring both under real throughput constraints.
- **Broad platform ownership:** search infrastructure (Elasticsearch/Algolia), analytics event tracking, third-party webhook integrations (Shopify, BlueTree), Vue.js frontend work, Laravel Dusk end-to-end tests, and internal admin tooling.

Software Developer II — *Benevity, Toronto, ON* January 2017 – April 2021

- **Architected and built new API services** as part of a small team; built the Grants Forms service from the ground up, including a customizable adapter for routing submissions to arbitrary receiving APIs.
- **Owned the team's weekly release process end-to-end:** triaged tagged Jira tickets, built and synced Docker images with AWS, ran the accompanying MSSQL scripts, coordinated with QA to validate the release, and stayed ready to roll back to a prior image if anything broke.
- **Introduced a test suite and TDD tooling** on a team recognized internally as the company's best-performing — regularly consulted by larger, less-efficient teams for advice on improving their own process.

Earlier Experience

Intermediate Web Developer, BID13 (2016–2017) — Drupal migrations, mentored junior developers.

Developer / Systems Analyst, Techcom (2014–2016) — Led client-facing development, coordinated a project team on secure financial applications.

Front-End / PHP Developer, 3conx & Jackpine (2013–2014) — Early-career web development.

Education

Algonquin College — Advanced Diploma, Computer Engineering Technology – Computing Science (2010–2013)